Regular maniplexes on the projective line

human_compression project

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1 Results recovered and common theory

This application recovers the principal results of Leemans–Toledo [1] from the common paper [2]. The source is the eleven-page author manuscript dated 15 May 2023. The claims recovered here are its rank restriction, Theorem 3.10, Proposition 4.2, Theorem 4.4, the two infinite constructions and Corollaries 4.6 and 4.8. We also correct the source's restricted list of possible face counts and several auxiliary statements. The corrections are distinguished from the source's results; no priority or new conjecture claim is made.

Write $L(q) = \mathrm{PSL}_2(q)$, $P(q) = \mathrm{PGL}_2(q)$, and $G_I = \langle r_i : i \in I \rangle$ for a marked group. A string marking is an ordered list of distinct nonidentity involutions generating G, with $r_i r_j = r_j r_i$ for $|i-j| > 1$. Its type lists the orders of adjacent products. The intersection property is not part of this definition. For rank four put $H = G_{\{0,1,2\}}$ and $K = G_{\{1,2,3\}}$.

We use the common paper's projective string theorem, classification when both end groups are proper, geometric identification and half-turn extension theorem [2, Theorems 4, 6, 8, 9 and Corollary 7]. These are proved there for string markings, so their use here does not require the intersection property as a hypothesis. In particular, the local S_4/A_5 classification is the exhaustive 432-triple computation in the common required program: generation and commuting endpoints are admission conditions, and the intersection property is verified afterward. The same program contains the complete local presentations, ten universal groups and literal correction witnesses used below. Its full cost is part of the common theory.

In addition we use the usual projective subgroup and torus theory declared in the common paper, Giudici's precise maximal A_5 and S_4 cases [3, Theorem 2.2(6),(8)], and Dirichlet's theorem for primes in coprime progressions. The final Suzuki observation uses the stated natural-action fact. We do not import an earlier version of this application's proof as an additional premise.

2 From marked groups to maniplexes

We first prove the correspondence between string markings and regular maniplexes. A maniplex has flags and fixed-point-free involutory adjacencies s_i, such that $s_i s_j$ is fixed-point-free for $i \neq j$, nonconsecutive adjacencies commute, and the adjacencies act transitively. An automorphism preserves every adjacency; a maniplex is regular when its automorphism group is transitive on flags. As in [1, Remark 2.1], a rank-three maniplex is called a map; this does not assert that its face poset is an abstract polytope.

Given a string marking of G, take its elements as flags and let s_i act by left multiplication by r_i. Nonidentity gives fixed-point-free adjacencies, distinctness gives fixed-point-free products of two different adjacencies, and generation gives connectedness. An adjacency-preserving permutation f commutes with all left multiplications. If $f(1) = h$, then $f(g) = gh$ for every $g \in G$; conversely every such right multiplication preserves the adjacencies. Thus the full colour-preserving automorphism group is the right regular group, isomorphic to G by taking inverse right multiplications. This proves equality with the full automorphism group, rather than merely the presence of a flag-transitive subgroup.

Conversely, in a connected maniplex an automorphism fixing one flag fixes all flags, by following coloured paths. Hence in a regular maniplex there is a unique automorphism taking a chosen base flag to each adjacent flag. Call these automorphisms r_i. They commute with every adjacency. Moving the base flag along a path shows that its terminal automorphism is the reversed product of the corresponding r_i; these therefore generate. The fixed-point-free and string conditions give distinct nonidentity involutions and the required nonconsecutive commutations. Changing the base flag conjugates the marking. This proves the correspondence up to marking-preserving isomorphism.

An i-face is an orbit of all adjacencies except i. In the regular model its number is the index of the corresponding generated subgroup. Reversal of the marking gives the dual maniplex, so an automorphism sending every r_i to r_{3-i} is a duality. The subgroup of even words has index at most two. For nonabelian simple G it must be G: otherwise G would have a nontrivial quotient of order at most two. The even adjacencies then act transitively, which is exactly nonorientability in the source's convention.

3 Classification and extension as corollaries

Corollary 1 (Leemans–Toledo)**.** *If $L(q)$ admits a string marking of rank at least four, then q is odd and the rank is four. Each end group is S_4, A_5 or $L(q)$. If both ends are proper, the marked object is uniquely the 11-cell for $q = 11$, of type $\{3, 5, 3\}$, or the 57-cell for $q = 19$, of type $\{5, 3, 5\}$. Both satisfy the intersection property and are self-dual.*

Proof. Apply the projective reduction and the classification when both end groups are proper in [2, Theorems 4 and 6] to $G = L(q)$. The possibility $\mathrm{PSL}_2(q) < \mathrm{PGL}_2(q)$ in the list of proper end groups does not arise for $G = L(q)$. The finite theorem leaves $L(11)$, $L(19)$ or S_5; simplicity excludes S_5. For odd q, the strictly increasing order $q(q^2 - 1)/2$ identifies the remaining field sizes as 11 and 19. The common geometric theorem identifies the two surviving marked groups and their dualities. Its subgroup indices give 11 vertices and 11 facets in the first case, and 57 of each in the second. Their facets are the hemi-icosahedron and hemi-dodecahedron, respectively, and their vertex figures are the dual local maps. This is precisely the source's Theorem 3.10, Proposition 4.2 and characteristic-two exclusion, together with the asserted rank bound. The small groups $L(2)$ and $L(3)$ are included in the common theorem's explicit exceptions check. □

Corollary 2 (Leemans–Toledo, Theorem 4.4)**.** *Let q be odd, choose $\alpha \in \{1, -1\}$ with $q \equiv \alpha \pmod 4$, and set $N = (q - \alpha)/2$. A rank-three string marking (r_0, r_1, r_2) of $L(q)$ whose first adjacent order is an odd divisor of N extends to rank four with the same group and the original map as its unique facet.*

Proof. The common indecomposability argument makes both old adjacent labels at least three. The case $H = L(q)$ of the half-turn extension theorem in [2, Theorem 9] supplies a new involution r_3 and proves all four generators distinct, their string commutations and generation. It preserves the first three generators, which generate $L(q)$. The correspondence in Section 2 therefore gives one facet, with exactly the original marked map. This uses the full odd-divisor range of the common theorem, not a finite list of fields. □

The same theorem also applies when H is a maximal proper subgroup. If $H < L(q)$ is maximal, its first adjacent order is an odd divisor of N, and $\mathrm{C}_H(r_0, r_1) = 1$, the torus half-turn lies outside H and the extended marking generates $L(q)$. The subgroup generated by the first three generators is H. We now prove the remaining details of these constructions.

4 A family with both end groups equal to the whole group

Theorem 3 (Leemans–Toledo, Lemma 4.5 and Corollary 4.6)**.** *For every prime $p \equiv 1 \pmod{12}$ there is a rank-four regular nonorientable maniplex with group $L(p)$ and both end groups equal to $L(p)$. Its*

first three generators have type $\{3, p\}$ *and give the source's marked map. There are infinitely many such maniplexes.*

Proof. Consider the projective classes of

$$R_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad R_1 = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}, \quad R_2 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad R_3 = \begin{pmatrix} 1 & -2 \\ 2 & -1 \end{pmatrix}. \tag{1}$$

The first three determinants are -1 and the last is 3. Since $4 \mid p-1$, -1 is a square. Choose ζ of order three; then $(2\zeta+1)^2 = -3$. Multiplying by a square root of -1 shows that 3 is a square too. Thus all four projective classes lie in $L(p)$. Their matrices have trace zero and nonzero determinant, and none is scalar, so they are nonidentity involutions.

Direct multiplication gives $r_0 r_2 = r_2 r_0$, $|r_0 r_1| = 3$, and

$$[R_1 R_2] = \left[\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \right]. \tag{2}$$

The latter has exact order p. The adjacent orders and the displayed matrices also show that the first three classes are distinct. Their powers supply all lower unipotent elementary matrices over $\mathbb{F}_p$; conjugation by R_0 supplies all upper ones. These generate $\mathrm{SL}_2(p)$ by elementary row elimination. To include the last diagonal step explicitly, let

$$u(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \quad v(t) = \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}, \quad h(t) = u(t)v(-t^{-1})u(t) = \begin{pmatrix} 0 & t \\ -t^{-1} & 0 \end{pmatrix}.$$

For $t \neq 0$ one has $h(t)h(-1) = \mathrm{diag}(t, t^{-1})$. The matrix $h(1)$ permits a row interchange when required, and the upper and lower matrices eliminate the other entries. Hence the first three projective classes generate $L(p)$.

Here $3 \mid (p-1)/2$, so the common extension theorem applies. Put $A = R_0 R_1$. The identity $2A - \mathrm{tr}(A)I = R_3$, together with the common dihedral-centralizer formula in [2, Lemma 1], proves that r_3 centralizes r_0, r_1. Both old adjacent pairs are noncommuting, so the extension theorem with $H = L(p)$ also proves that this fourth involution repeats none of the old generators. Equivalently, it is the unique torus half-turn in Corollary 2. The other end contains $r_1 r_2$ of order $p \geq 13$. Corollary 1 makes it S_4, A_5 or $L(p)$, and the first two have no element of order p. Both ends are therefore full.

The construction agrees with the actual source marking, not merely its group order. Choose $a \in \mathbb{F}_p$ with $a^2 = -1$. The three source matrices are

$$S_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad S_1 = \begin{pmatrix} a & 0 \\ 1 & -a \end{pmatrix}, \quad S_2 = \begin{pmatrix} -a & 0 \\ 0 & a \end{pmatrix}.$$

For $C = \mathrm{diag}(-a, 1)$ their conjugates are respectively $-aR_0, aR_1, aR_2$. Thus the ordered projective triples are conjugate. Uniqueness of the nonidentity common-centralizer element identifies the fourth marker as well. The group–maniplex correspondence gives one vertex and one facet. Simplicity proves nonorientability. Finally, Dirichlet's theorem supplies infinitely many primes in the progression, and the strictly increasing orders $p(p^2-1)/2$ distinguish the groups. □

5 Proper alternating and symmetric ends

Theorem 4 (Leemans–Toledo, Proposition 4.7 and Corollary 4.8)**.** *For every prime* $p \equiv \pm 1 \pmod{20}$, $p \neq 19$*, there is a rank-four regular nonorientable maniplex with group* $L(p)$*, one vertex and* $|L(p)|/60$ *hemi-dodecahedral facets. Its end groups are* $L(p)$ *and* A_5*. There are infinitely many such maniplexes.*

Proof. Giudici's theorem gives a maximal A_5 subgroup H for every prime in the stated range. Choose the common local marking of type $\{5, 3\}$ with Petrie order five. Its first pair generates D_{10}. The centralizer of a 5-cycle in A_5 is its cyclic group of order five; the dihedral reflection inverts it, fixing only its identity. Thus $C_H(r_0, r_1) = 1$.

Let $p \equiv \alpha \pmod 4$, $\alpha \in \{1, -1\}$. If $p \equiv 1 \pmod{20}$ then $\alpha = 1$ and $10 \mid (p-1)/2$; if $p \equiv -1 \pmod{20}$ then $\alpha = -1$ and $10 \mid (p+1)/2$. The first adjacent order five is therefore an odd divisor of the even torus order $N = (p - \alpha)/2$. The extension theorem for maximal proper subgroups gives a fourth involution outside H, and the resulting marking generates $L(p)$ with H as the subgroup generated by the first three generators.

If its opposite end were proper, Corollary 1 would give $p = 11$ or 19. Neither is among the present parameters. The opposite end is full, giving the asserted face counts. The marking of H identifies the facets as hemi-dodecahedral. At $p = 19$ the same extension construction still makes a string marking, but this argument does not force the opposite end to be full; this is why the source excludes that value from its family assertion. Simplicity and Dirichlet complete the nonorientability and infinitude claims. □

Theorem 5 (Correction: the omitted symmetric-end family)**.** *For every prime $p \equiv \pm 1 \pmod{24}$ there is a rank-four regular nonorientable maniplex with group $L(p)$, one vertex and $|L(p)|/24$ facets. Its facet marking has type $\{3, 4\}$ and group S_4, and its other end group is $L(p)$. There are infinitely many such examples.*

Proof. Giudici's maximal-subgroup theorem gives a maximal S_4 subgroup H for these primes. Use its common local marking of type $\{3, 4\}$ with Petrie order three. The first pair generates an S_3 fixing one point in the natural degree-four action. An element centralizing this S_3 must preserve that fixed point; on the other three points it lies in the centre of their full symmetric group and is therefore trivial. Hence $C_H(r_0, r_1) = 1$.

For $p \equiv 1 \pmod{24}$, $N = (p-1)/2$ is divisible by twelve; for $p \equiv -1 \pmod{24}$, $N = (p+1)/2$ is divisible by twelve. The first adjacent order three therefore satisfies the hypothesis of the extension theorem for maximal proper subgroups. The resulting fourth involution lies outside H and generates $L(p)$ with it. These primes satisfy $p \geq 23$, so the opposite end cannot be proper by Corollary 1. Its fullness gives the face counts; simplicity gives nonorientability, and Dirichlet gives infinitude. □

Consequently the end-group and face-count alternatives, up to duality, are exactly

end groups	(vertices, facets)
G, G	$(1, 1)$
G, S_4	$(1, \lvert G\rvert/24)$
G, A_5	$(1, \lvert G\rvert/60)$
two proper ends	$(11, 11)$ or $(57, 57)$.

This lists the possibilities for end groups and counts, not all marked isomorphism classes. In both cases with exactly one proper end group, the maniplex is not self-dual because the two counts differ. When both end groups equal G, the maniplex can be self-dual.

6 Literal corrections and exact witnesses

The source's introduction and conclusion omit the S_4 alternative, although its Theorem 3.10 and Remark 4.1 permit it. Theorem 5 gives a uniform correction. Independently of that uniform argument, the common program supplies a literal degree-24 marking of $L(23)$ of type $\{3, 4, 12\}$, with whole order 6072 and end orders 24, 6072. It checks distinct involutions, all distant commutations, generation and the S_4 identification. This gives one vertex and 253 facets directly.

For the source page 9 self-duality sentence, use the following marking of $L(13)$ in degree fourteen:

$$r_0 = (3\ 9)(4\ 10)(5\ 11)(6\ 12)(7\ 13)(8\ 14),$$
$$r_1 = (1\ 4)(2\ 14)(3\ 9)(5\ 13)(6\ 11)(7\ 12),$$
$$r_2 = (1\ 2)(3\ 7)(4\ 6)(8\ 14)(9\ 13)(10\ 12),$$
$$r_3 = (1\ 2)(4\ 14)(5\ 13)(6\ 12)(7\ 11)(8\ 10).$$

The common program verifies whole and end orders 1092, type $\{3, 7, 3\}$, and all four conjugacies $r_i^d = r_{3-i}$ for

$$d = (1\ 3)(2\ 9)(4\ 7)(5\ 8)(6\ 12)(10\ 11)(13\ 14).$$

Thus conjugation by d induces a reversal automorphism, and the corresponding maniplex, whose two end groups equal $L(13)$, is self-dual. The obstruction from unequal face counts applies when exactly one end group is proper.

The source's Lemma 3.3 asserts involution-centralizer maximality too broadly. The common program verifies in $L(5) \cong A_5$ that such a centralizer is a four-group properly contained in A_4. Our argument uses the common-centralizer theorem and torus normalizers, without assuming maximality of every involution centralizer. The broad uniqueness assertions in source Lemma 3.4 are not used here. Likewise, Lemmas 3.7 and 3.8 must concern natural subfield embeddings, not arbitrary abstract isomorphisms: $A_5 \cong L(5)$ and $S_4 \cong P(3)$ occur as allowed proper ends. Finally, the source first correctly calls the local map of type $\{5, 3\}$ hemi-dodecahedral, then calls its facets hemi-icosahedral. The actual marking has been preserved here and the name made consistent.

7 Remaining scope and verification

The source's Suzuki observation follows from the following fact about its natural action: every involution in a simple Suzuki group fixes a unique point, and commuting involutions have the same fixed point. At rank four, r_0, r_2, r_3 must fix the same point, and then so does r_1 because it commutes with r_3. Every later generator commutes with r_0 and fixes that point as well. This contradicts generation of the transitive group. The nonsimple group $\mathrm{Sz}(2)$ cannot be generated by involutions, since its quotient C_4 cannot be so generated. This observation is included with its original attribution; no additional Suzuki classification is claimed.

The final PGL remarks and conjecture about four families in [1, Section 5] are treated separately. No complete enumeration of rank-four PSL markings is claimed here. Unused auxiliary subgroup multiplicities and uniqueness clauses are not presented as separately recovered theorems. These qualifications leave unchanged the list of principal results stated in Section 1.

The common finite program is a required mathematical component. These finite verifications rely on GAP's algorithms for finite groups and finitely presented groups. Finite samples of the positive families are useful diagnostic checks; the proofs above establish their full prime ranges and do not infer them from samples. The displayed conjugacy identifies our generators with those in the source for every admissible prime. The generic extension now has a separate Lean certificate proving actual tuple injectivity, string relations, generation and preserved prefix under its stated hypotheses. Earlier Lean certificates prove scoped centralizer and rank interfaces. They do not formalize torus existence, Dickson or Giudici classification, the finite universal table, Dirichlet, the maniplex correspondence or every theorem of this paper.

References

[1] D. Leemans and M. Toledo, *Maniplexes with automorphism group* $\mathrm{PSL}(2, q)$, author manuscript dated 15 May 2023, eleven pages.

[2] human_compression project, *Common theory for projective string markings*, common Topic E theory, accompanying `umbrella.tex` and complete `required_finite.g`. Precise theorem labels and final edition hashes are bound in the joint proof map.

[3] M. Giudici, *Maximal subgroups of almost simple groups with socle* $\mathrm{PSL}(2, q)$, arXiv:math/0703685v1, 2007, Theorem 2.2, physical and printed page 4.